

1 Application Generator

Applications may be generated in an automated way by the application generator. Automation refers in particular to standard features that are used in many applications (above all editing dialogs) and that may easily be standardized for this reason.

The application generator receives all its information from the repository or the files exported from there. Consequently, it is reasonable to edit all affected data in the repository before a new application is generated.

Each new application can and should be created using the generator. If the entries available in the repository are identified as being incorrect after the generation, they should be modified first before generating a new application.

Each application may still be changed later. Some changes can be made at runtime using editors and others can only be performed by directly editing the configuration files. If possible, modifications should always be made via the repository, as in this case they can also be used for other applications.

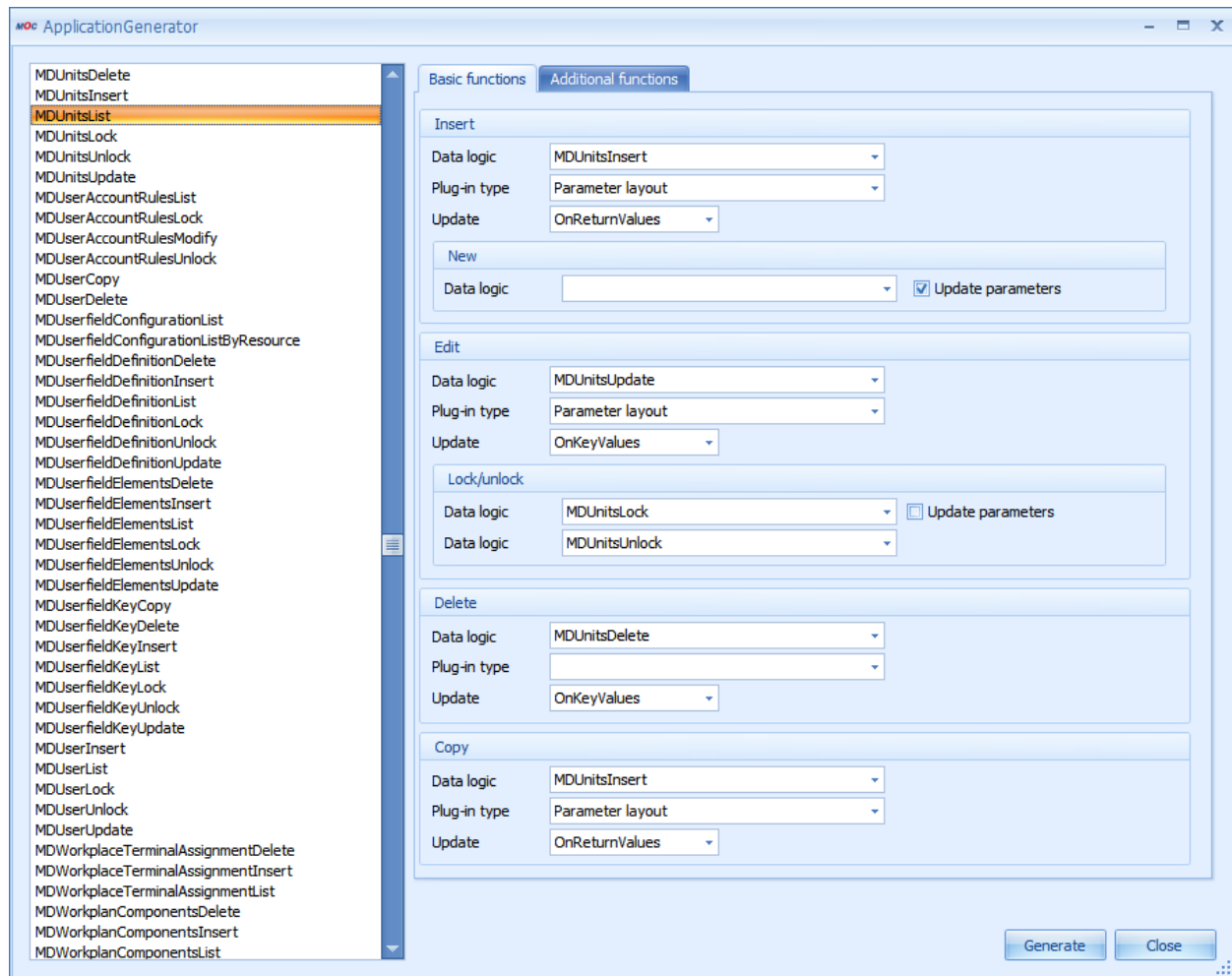
A generated main application always includes:

- a (main) data source
- a selection panel (Plug-in = ParameterLayout, Configuration file = LayoutPanel.config)
- A grid application to show the data for the data source (Plug-in = Grid, Configuration file = GridMainGrid.config)
- A detail application to show the data records selected in the grid (plug-in = Layout, configuration file = LayoutMainDetail.config)
- 0-n toolbar buttons to start editing functions (configuration file = ApplicationCommandLinkCollection.config)
- 0-n (sub-)applications to implement the editing functions

If additional editing applications are generated, they also include data sources and, if necessary, detail applications. Moreover, they can also be edited and saved at runtime, just as it is the case for "normal" main applications.

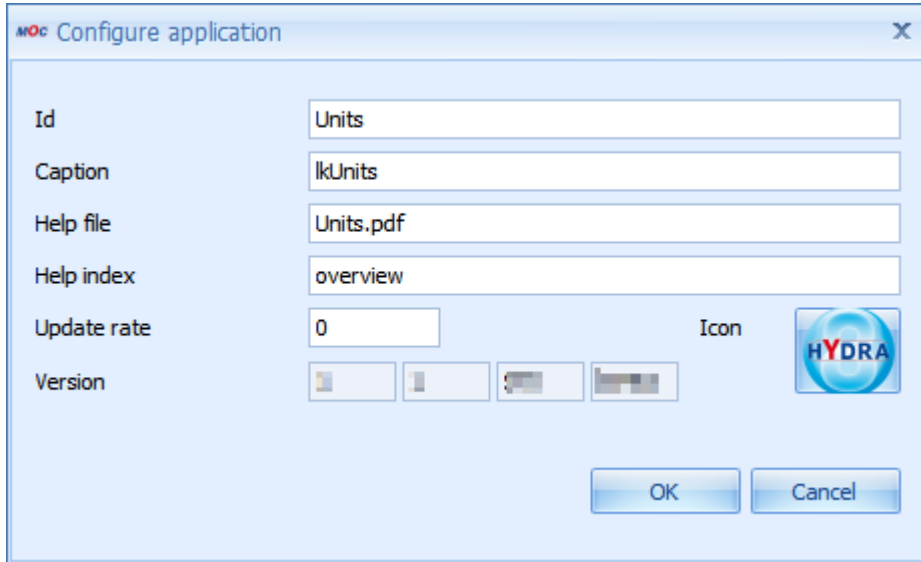
1.1 Instructions

The application generator is available if the MES Development Suite is enabled and can be started by the shortcut _cgenapp in the quick launch bar or by using the main menu (MES Development Suite → Generate application).



- Data logic has to be selected first from the list on the left-hand side.
 - This list includes all data logic exported from the repository into the files in <MOC>\resources\data\services\xxx.DataLogic.xml. If a specific entry is missing in the list, it should be checked whether or not the data have already been exported.
 - The selected data logic is then used as main data source within the main application (for the grid as well as for the detail application).
 - Normally, main data logic has the extension "List" or "Overview".
 - The selection process is simplified by "Incremental Search", i.e. if the first letters of the data logic name are entered, the matching entry will be selected automatically.
- Once data logic has been selected, most fields on the right-hand side of the generator are automatically assigned default values.
 - This pre-assignment refers to default values matching in many cases but certainly not in every case. Therefore, the field assignment has to be checked thoroughly as to whether it meets the requirements of the application to be generated (see the description of the fields).
 - At the moment, the basic functions Insert, Copy, Update and Delete are automatically assigned as editing functions. If the relevant service is available, the corresponding field will be assigned automatically in the generator.
- Besides the standard editing functions, up to five further editing functions can be defined on the tab "additional functions" (e.g. Activate, Deactivate, etc.). No additional editing functions will be created if nothing is edited here.

- A dialog to configure the main dialog opens by clicking the "generate" button. This dialog is already assigned default values within the repository, if data logic has been configured correctly. However, the relevant specifications may also be entered manually.



- Once this dialog has been closed, the main dialog and, if necessary, editing dialogs of the application are generated and directly opened.

1.2 Description of the fields

The individual editing functions can be edited on the right-hand side of the generator. As already mentioned, the default functions are "insert", "copy", "edit" and "delete". Up to five further functions may be defined on the second tab. An editing application is generated automatically for each editing function for which data logic has been selected.

The following specifications can be made for each editing function:

Data logic

Data logic selects the service that is respectively started/used for the editing of data.

An editing application is generated automatically, once data logic has been selected. If the data logic name is empty, the created application will not include the editing application.

Data logic (New), which is executed before the actual insert process, may be defined additionally for "insert". Data logic for the new service is automatically assigned to default values when selecting list data logic, provided the relevant service is available.

Additional data logic may be specified for "update". This data logic is executed before the actual "update" process before (Lock) or after (Unlock). Data logic for Lock/Unlock is also assigned default values when selecting list data logic, provided the relevant services are available.

Most applications do not have a separate copy service. If, however, a copy function is still to be used, the insert dialog will normally be used and its fields will be assigned to default values. If a copy service exists, it will be assigned by default. Otherwise, the insert service is assigned automatically. Data logic has to be deleted here if no copy function is required.

Plug-in type

The plug-in type specifies which application plug-in is used for editing data.

Any application plug-in may be selected for each editing dialog. The only condition: the plug-in has to implement `IParameterContainer`. The selection list only includes appropriate plug-ins. The plug-in `ParameterLayout` is always set by default.

It is also possible to NOT select a plug-in. Consequently, a dialog without plug-in is created automatically. However, this dialog will not be opened by clicking the relevant toolbar button. Then the application folder includes a configuration that may be edited. Example: "standard delete" without dialog.

Update (UpdateSourceMode)

Specifies how data are to be updated in the main application, once the editing function has been executed. You may choose from the following options:

- All: all data of the main application are completely refreshed (taking the values entered in the selection panel into account). This is required, for example, for separate deletion and copy dialogs. As they do not allow for the affected data records to be determined specifically and requested only.
- OnReturnValues: the called service returns values that are used to request only the concerned data. This may be the case, e.g. for "Insert". In this context, it is often the case that a new key field is generated that is required to request the new data record.

Please note: However, this setting is not reasonable if the called service does not return any fields that are suitable for a unique identification of the affected data record. This setting is set by default for "inserting" and might be changed, if necessary.

- OnKeyValues: the key fields of the list service are used to request only the affected data. This is, for example, the ideal procedure for the "update".

Please note: In this case, the service of the main application and the editing service should have the same key fields; otherwise it does not work.

- OnScript: for future use

This configuration should be checked thoroughly. As subsequent changes can only be made manually in the main configuration file of the editing dialog (e.g. `EditOrders\Insert\Insert.config`, section: `UpdateSource`).

Update parameter

This entry specifies whether or not the return value of a service is to be used to replace the original parameters. These new parameters will then be used during the course of the process.

Example: Calling the new service before the "insert" is normally used to request data that will then be assigned by default in editing dialogs. Therefore, the data provided by this service are required during the course of the process. Consequently, it is reasonable to set the check box to "true". For this reason, this is the default value assigned for "New".

Name

Name of the editing function. This value has already been assigned as the default value for the basic functions and can only be assigned individually for the additional functions. This name is used for the following positions:

- The editing application will be named like that as a part of the main application. Consequently, the folders included in the configuration also have this name. For this reason, the name has to be clear and unique within the main application.
- Ik + Name = labeling of the created toolbar button
- The command link on which the toolbar button is based is named like that.

Small/large

Icons for the application may be selected by image dialogs. These icons are used for the following positions.

- Small icon (16X16): presented top left of the editing application; possibly also in the toolbar.
- Large icon (32X32): might be displayed in the toolbar (both icons should be defined as the ribbon defines whether the large or the small icon is shown)

Transfer of parameters (ParameterProcessingMode)

Please note! This configuration can neither be set by the generator nor by a wizard!

The configuration `ParameterProcessingMode` specifies if and how parameters from the calling main dialog are transferred to the editing dialog. You may choose from the following options:

- `NoParameter`: no parameters will be transferred
- `SingleParameter`: obligatory transfer of the selected grid row; an error message occurs if no row is selected.
- `OptionalSingleParameter`: optional transfer of the selected grid rows; no error message
- `MultiParameter`: obligatory transfer of one or several grid rows; an error message occurs if no row is selected.
- `OptionalMultiParameter`: optional transfer of one or several grid rows; no error message.
- `SingleSelectionParameter`: transfer of the selected grid row. If no row is selected the values from the selection panel will be transferred. If no values are transferred from there --> error message.
- `OptionalSingleSelectionParameter`: transfer of the selected grid row. If no row is selected the values from the selection panel will be transferred. No error message.
- `SelectionOnlyParameter`: the values from the selection panel are transferred. If no values are transferred --> error message.

- OptionalSelectionOnlyParameter: the values from the selection panel are transferred. No error message.

As a part of the generation process, the following values are always set automatically for editing applications:

- Insert: NoParameter
- Update: SingleParameter
- Copy: SingleParameter
- Delete (including message box, without editing dialog): MultiParameter
- Delete (including editing dialog): SingleParameter

At the moment, these values can only be changed manually and directly in the configuration (e.g. EditOrders\Insert\Insert.config, section: ParameterProcessingMode).